Requirements and Acceptance Criteria

This specification defines 16 functional and 10 non-functional requirements for the MathVerity MVP, their priorities, acceptance conditions and associated checks. Skill status rules distinguish assisted performance from independent results. Both Kazakh-language and Russian-language interfaces are mandatory for the MVP.

Must means mandatory for the MVP; Should means desirable after the mandatory core. These requirements describe the target MVP, not completed implementation. No runnable application or executable prototype is included in the current deliverables. Tests T01–T26 are planned and are not claimed to have passed.

Functional Requirements

Table 1. Functional requirements

ID Priority	Requirement	Acceptance criterion	Module and test
FR01 Must	Family account and roles	A parent creates a child profile; sign-in creates a server session. A learner cannot elevate their role by modifying a request.	Identity T01
FR02 Must	Parent–child linking	A one-time code expires after 15 minutes and is created by an authenticated parent for their own profile. Reused or unauthorised codes are rejected; links can be revoked.	Family T02
FR03 Must	Catalogue and lesson	Learners can open published topics only. Each has a goal, example and skills. Drafts are unavailable to the learner role.	Content T03
FR04 Must	Practice and answer checking	Reference-equivalent answers are accepted for every supported type. Zero denominators and arbitrary code are rejected. The server determines correctness.	Assessment T04
FR05 Must	Error diagnosis	A diagnostic question stores the selected step and a possible error code. One error is not labelled a proven misconception.	Assessment T05

ID Priority	Requirement	Acceptance criterion	Module and test
FR06 Must	Stepwise hints	Practice offers up to three levels in Kazakh or Russian. The server checks the attempt and records assistance. If AI is unavailable or its output is rejected, an internally cross-reviewed template in the selected language is returned.	Tutor T06
FR07 Must	Assessment without hints	The server creates a new variant in independent mode. Its hint endpoint returns 403; the first answer is recorded once.	Assessment T07
FR08 Must	Explainable skill status	A versioned rule calculates status from independent attempts. The response includes supporting evidence, item count and date. Mastery is not declared when evidence is insufficient.	Progress T08
FR09 Must	XP and achievements	The first completed independent item earns 10 XP, plus 5 for a correct first answer. Repeating the same item instance adds nothing. XP does not affect diagnosis.	Gamification T09
FR10 Must	Parent report	Only linked children are shown, with skills, assistance, independent results, recency and recommended review. Another family's child_id is inaccessible.	Reporting T10
FR11 Must	Content authoring	Only the two developers hold internal authoring rights. The other developer must record a review of the answer, diagnostic steps, hints and both language versions before publication. Editing creates a new version; prior attempts retain their original version.	Content T11
FR12 Should	Next-item recommendation	Selection considers errors, prerequisites and recent variants. An empty bank returns an explanation rather than an infinite loop.	Progress T12
FR13 Must	Visible activity	The timer counts only a visible window with heartbeats and recent interaction. A gap over 60 seconds stops accumulation. Reports label this as estimated time within the platform.	Reporting T13
FR14 Must	Family data controls	Parents can revoke links and request export or deletion. Active APIs no longer return deleted profiles. Backup deletion is documented separately.	Family T14

ID Priority	Requirement	Acceptance criterion	Module and test
FR15 Should	Delayed review	A new independent variant is offered 7 days after the first positive signal, with an indication that delayed review is still pending.	Progress T15
FR16 Must	Kazakh and Russian support	Core learner and parent flows, tasks, hints, feedback and reports are available in both languages. Switching language preserves the attempt and scoring. The same answer and evidence give the same result in either language.	UI / Content / Tutor T26

Nonfunctional Requirements

Table 2. Nonfunctional requirements

ID Priority	Area	Acceptance condition	Test
NFR01 Must	Access control	The server rejects every operation prohibited by the role matrix, including access to another family's child_id. A 403/404 response exposes no child data.	T16
NFR02 Must	Assessment reliability	100% agreement with reference results for supported types on at least 200 boundary and equivalent-answer examples. One developer prepares the cases; the other independently checks the expected results. The LLM is not the reference evaluator.	T17
NFR03 Must	API latency	On a 2-vCPU/4-GB test environment using the verification profile: 50 virtual users, 10 minutes after warm-up; ordinary API p95 ≤500 ms and unexpected errors <1%. AI is measured separately.	T18
NFR04 Must	AI failure	An 8-second timeout; a fallback response within 9 seconds. Independent items and reports continue working with AI disabled.	T19
NFR05 Must	Input protection	Parameterized SQL, length and type validation, no eval, HTML escaping and a 16-KiB learning JSON limit. Negative request tests pass.	T20
NFR06 Must	Data minimisation	The LLM receives only internally reviewed item context, a numeric answer, skill/error codes and the requested language. No names, contacts, family identifiers or children's free text are sent; outgoing requests are inspected.	T21

ID Priority	Area	Acceptance condition	Test
NFR07 Must	Accessibility	Core flows support a keyboard, visible focus and aria-live results. No horizontal scrolling at 360 px. Text contrast ≥4.5:1 [T14].	T22
NFR08 Must	Recovery	MVP target: restore a test copy from a daily backup within 60 minutes, with at most 24 hours of data loss. A separate restore test is required.	T23
NFR09 Must	Observability	Errors carry a request_id. Logs contain no passwords, cookies or complete child text. AI latency, fallback and error counts are visible.	T24
NFR10 Must	Reproducibility	Dependency versions and migrations are pinned. A clean environment runs the target MVP using verified README instructions, including static-hint mode without an AI provider.	T25

Skill Status Rules

Rule version: 1.1. The assessment unit is a learner–skill pair. The rule version, supporting evidence and calculation time are stored. Recency is checked first: if independent attempts exist but the latest is more than 14 days old, the status is “new assessment needed”. If no attempts exist, the status is “insufficient evidence”.

A current assessment uses first answers on new item instances in independent mode over the last 14 days. Repeated instances of the same template with identical parameters are excluded. Sufficient evidence requires at least five items, at least two different templates, two calendar dates, a completed transfer item and an answer to a diagnostic justification question. The pilot time zone is fixed in the protocol. An incorrect justification remains evidence: it does not turn poor performance into “insufficient evidence”.

If volume or variety is insufficient, the status is “insufficient evidence”. Otherwise, first-answer accuracy on numeric or selected-response items below 60% means “review needed”; 60% to below 80% means “developing”. “Confirmed on the current item set” requires accuracy of at least 80%, a correct transfer answer and a correct structured justification. If accuracy is at least 80% but transfer or

justification is incorrect, the status is “developing”. The separate diagnostic justification question does not increase the denominator for main-item accuracy.

The 80% threshold and the other conditions are design hypotheses. Both developers cross-review the rules and manually check example classifications; user interviews and a pilot must evaluate their usefulness. A status is not a probability of understanding, identity verification or proof that no external help was used. Free-form explanations may support consented manual team review, but do not replace deterministic criteria. The LLM does not determine the final status. A new rule version requires recalculation and display of its version number.

Access Matrix

The target users are learners and parents. Internal content authoring is a restricted capability of the two project developers, not an external expert role. Development or authoring privileges do not automatically grant access to identifiable family results.

Table 3. Access rights by role

Resource	**Learner**	**Parent**	**Internal content author (project developer)**
Attempts	Own attempts only	Aggregates for linked children	No personal attempt access
Hints	Own practice	No impersonation of the child	Template previews without child data
Reports	Own progress	Verified links only	Synthetic or anonymised examples only
Content	Published versions	Published versions	Draft and publish after the other developer’s recorded review
Family data	Own alias and settings	Links, consent, export/deletion requests	No access

Consent, AI eligibility for the selected age group, retention periods and provider conditions must be approved before a real pilot involving children.

Requirements Verification Plan

The target MVP is checked at four levels: unit tests for mathematical and diagnostic logic; integration tests with real PostgreSQL; API and access-control tests; and end-to-end learner, parent and internal authoring flows. T01–T26 correspond to the acceptance criteria. Both fullstack developers implement and test their workflows across the client and server and cross-review each other's work. A test plan does not establish that tests have passed.

T01–T03 cover roles, linking and publication. T04–T05 cover answer normalisation and diagnostic steps. T06–T07 cover hints, fallback and the server-side hint prohibition in independent mode. T08–T09 cover skill statuses and duplicate XP awards. T10–T11 cover access to another family's data and published-version immutability. T12–T15 cover recommendations, visible activity, deletion and delayed practice. T16–T25 cover security, the reference evaluator, load, AI failure, invalid input, leaks, accessibility, restore, logs and reproducibility. T26 covers the bilingual MVP.

T26 executes registration and linking, catalogue, practice, independent assessment, hint fallback and parent reports in both Kazakh and Russian. It checks complete published wording without untranslated keys or mixed-language fallback, preserved state after switching language, identical scores and statuses for paired items, keyboard operation and the 360-px layout in each language. A published item must have both language variants; translations are not counted as additional mathematical items.

Required negative cases include parent A requesting child B, a learner calling an author endpoint, an expired linking code, repeated submissions with identical and different payloads, hints for an independent item, a zero denominator, an empty answer, excessive input length, SQL/HTML/prompt injection, AI failure, duplicate reward delivery and concurrent submissions. The item API response must not expose the correct answer or private rubric.

T11 also rejects publication without the other developer's completed review, self-approval, a missing language variant and unresolved answer or hint defects.

The review record includes author and reviewer identifiers, content version and timestamp. The 120-item target covers eight skills with 15 distinct items per skill; all texts, translations, reference answers, diagnostic choices, templates and pilot forms are prepared by the project team.

T08 must include boundary cases: five complete incorrect attempts yield “review needed”, not “insufficient evidence”; 4/5 correct with an incorrect justification yield “developing”; 4/5 correct with correct transfer and justification yield “confirmed”; four attempts yield “insufficient evidence”; and a latest attempt older than 14 days yields “new assessment needed”. Variety and date conditions must also be checked in every case.

Load Test Conditions

The Go API and PostgreSQL share a test environment with 2 vCPUs and 4 GB RAM. A separate load generator runs on the same local network. The test data must contain 100 synthetic learner profiles, 120 published items and 10 verified family links. Warm-up lasts 2 minutes, followed by 10 minutes of measurement. There are 40 virtual learners and 10 virtual parents, each issuing sequential requests with a 2-second pause between actions.

The learner cycle is catalogue → new item → answer submission; the parent cycle requests a linked child’s report. Successful responses and expected validation failures are counted separately from unexpected 5xx responses and timeouts. Target p95 latency and error rates apply to ordinary API requests and are also recorded per endpoint. Hints using an external LLM are measured separately so that provider latency is not hidden in an overall average. Record versions, configuration, SQL indexes, duration, request counts and actual throughput.